

Evolution of Academic Performance in Spain

Airam Fernández, Amaiur Milla, Julen Corera

Entire project

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01

Recap

Deliverable 1

Selected data 2013 - 2024

Datasets AC

- Educational attainment

lower secondary, upper secondary and higher education

ages:  25-64, 25-34 and 55-64

- Suitability rates

ages between 12 and 15

Dataset F1

- Labour-market indicators

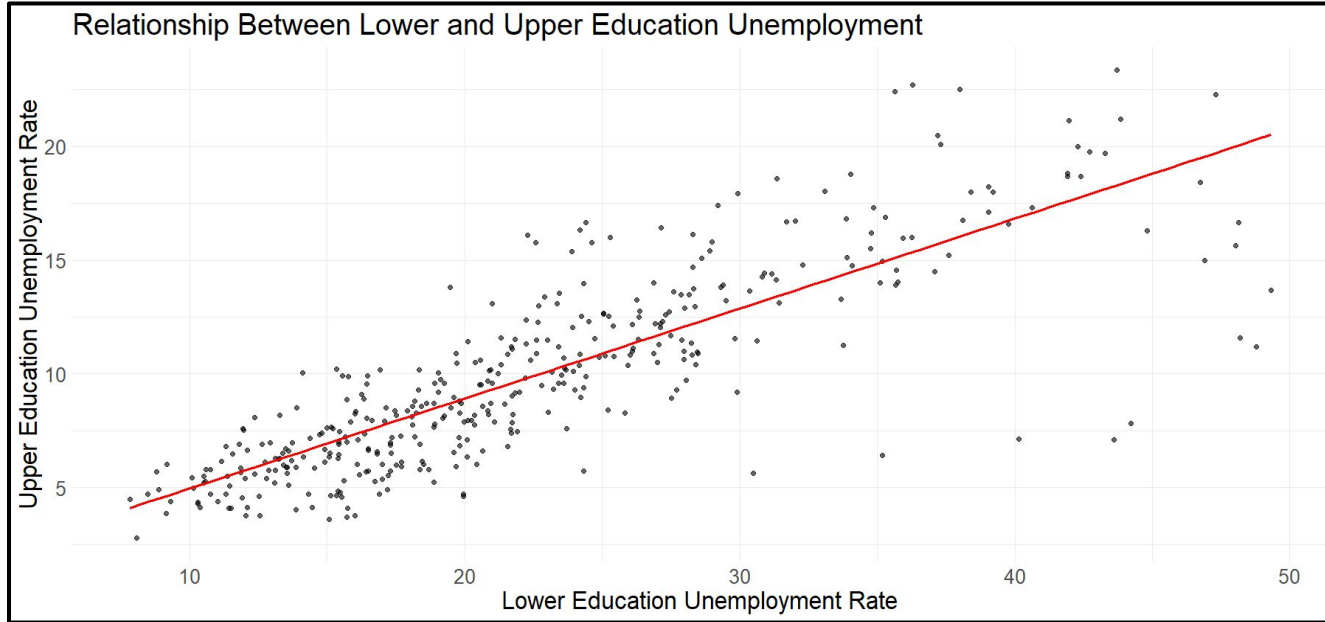
rates  unemployment, employment, activity

lower education, middle education and higher education

ages  25-64 and 25-34

Combination of Autonomous Community, sex and year

Upper Unemployment Trend



suggesting that unemployment patterns are connected across education groups

02

Linear/Logistic

Deliverable 2

Linear Regression recap

- Target variable: `low_unemployment_rate`
- Selection: backward stepwise (AIC/BIC + VIF)
- Result: significant model (F-test) — key predictors: year, sex, upper secondary education
- Validation: 70/30 split, reasonable predictions

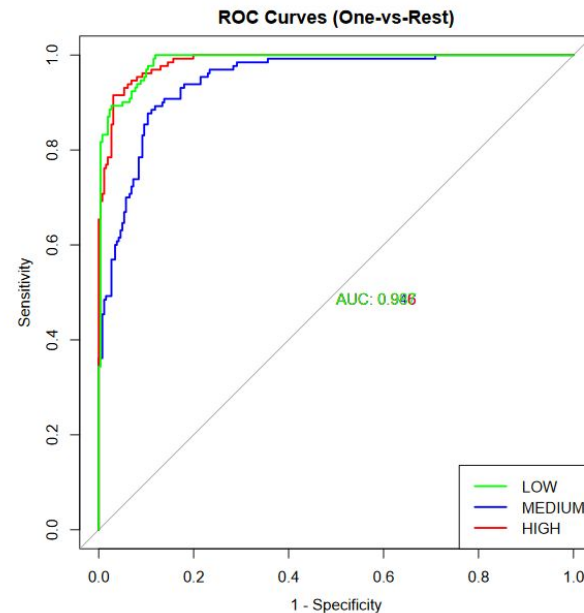
Logistic Regression recap

1. Goal and data

- **Target:** Predict employment level (LOW / MEDIUM / HIGH)
- **Feature Selection:** Optimized using VIF, Wald tests (p-value), AIC, and BIC

2. Key Results

- **Overall Accuracy:** 87.2%
- **AUC per class:** Low: 0.986 | Medium: 0.946 | High: 0.987



03

PCA

Deliverable 3

Why we use PCA?



Objective: Reduce 26 correlated variables.

- Identify hidden structures



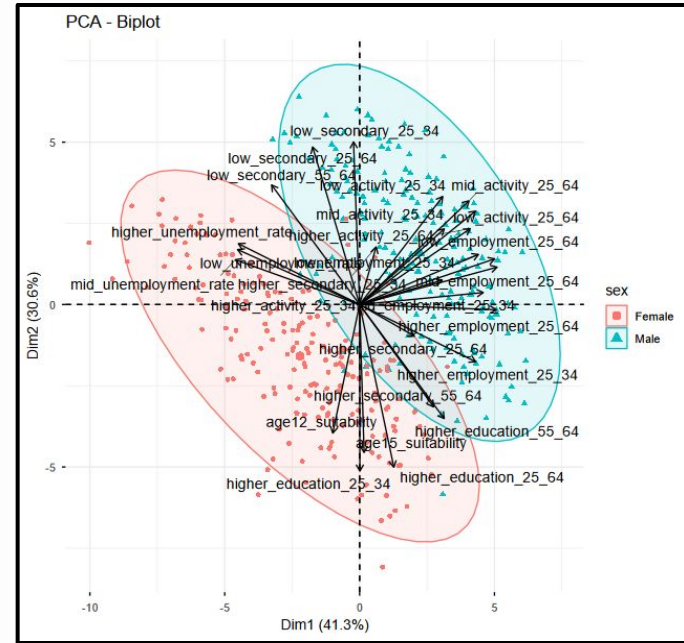
Key: 71.9% of explained variance (Dim 1 + Dim 2).



Standardized data (scale.unit = TRUE)

PCA: Interpretation of the Dimensions

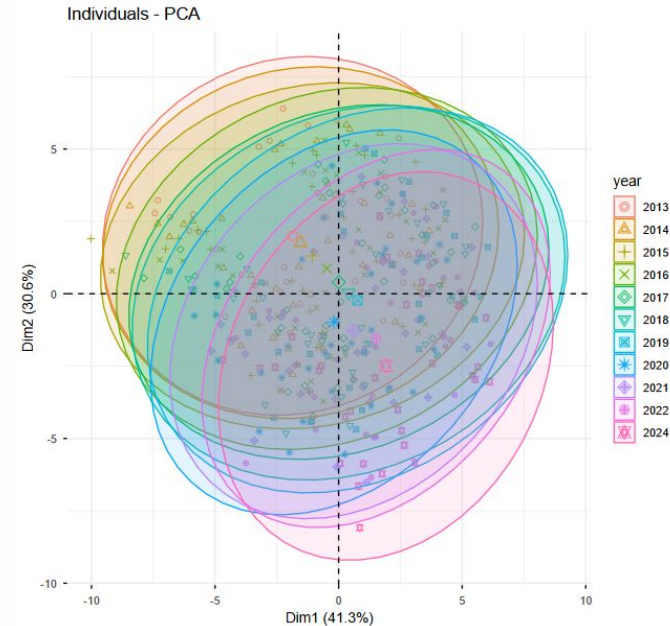
- Dim 1 (41.3%): Labour Market (Employment and activity rate).
- Dim 2 (30.6%): Education (Attainment and suitability).



Key Findings: Trends & Outliers

- **Time Trend:** Diagonal shift (2013 to 2024) showing progressive socioeconomic improvement.
- **Outlier Detection:** Female observations from Ceuta and Melilla isolated due to atypical labor profiles.

	autonomous_community	sex	year
322	Ceuta y Melilla	Female	2015
214	Ceuta y Melilla	Female	2018
286	Ceuta y Melilla	Female	2016
358	Ceuta y Melilla	Female	2014
394	Ceuta y Melilla	Female	2013



04

CA

Deliverable 3

Contingency table



autonomous_community and *low_unemployment_rate*



Using quantiles



Low, Medium, High

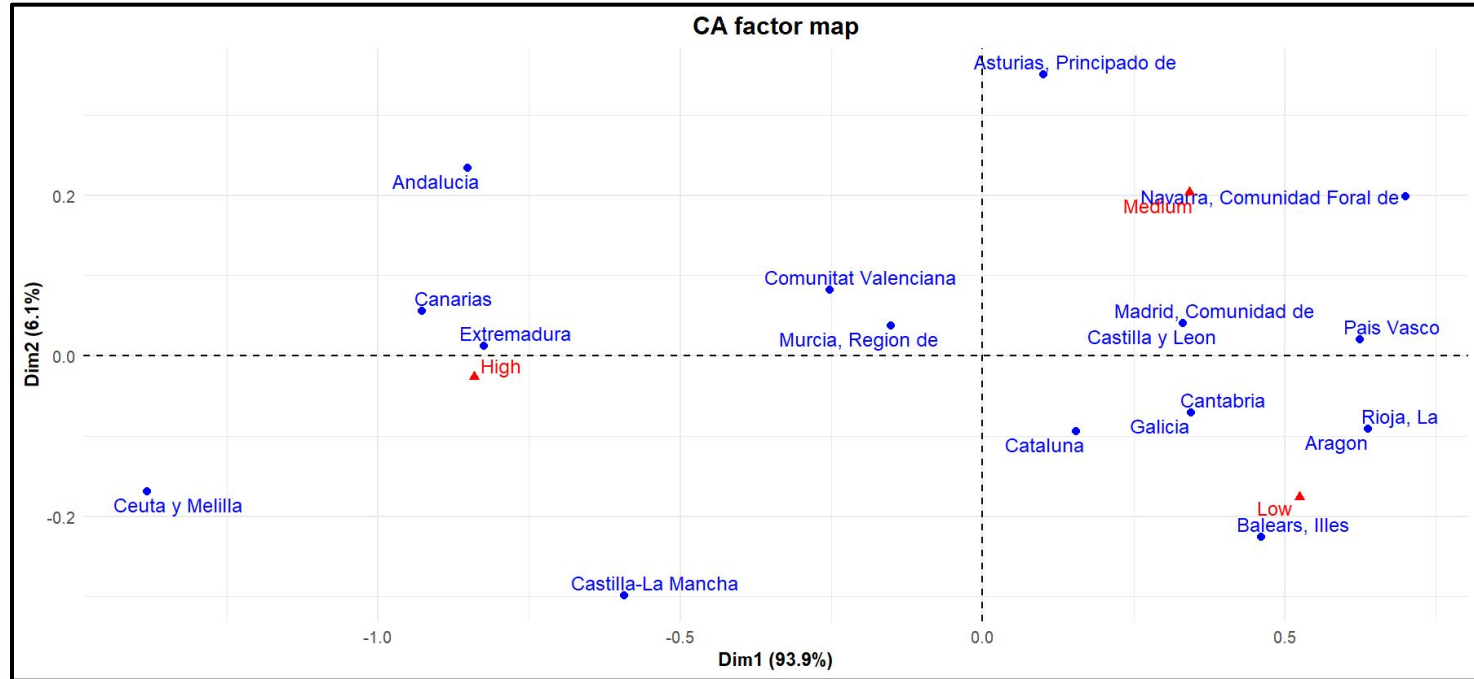


Small p-value

Pearson's Chi-squared test

```
data: unemployment_table  
X-squared = 154.15, df = 34, p-value < 2.2e-16
```

Factor map



Standardized residuals

	autonomous_community	associated_level	max_residual
Andalucia	Andalucia	High	3.1132837
Aragon	Aragon	Low	1.7600101
Asturias, Principado de	Asturias, Principado de	Medium	1.3888321
Balears, Illes	Balears, Illes	Low	1.7600101

Regions associated with unemployment levels

Unemployment level and Autonomous Community are not independent

05

Clustering

Deliverable 3

Objective



Identify profiles

Can we group observations into distinct socioeconomic profiles?



Compare methods

Do K-means and hierarchical clustering produce consistent results?

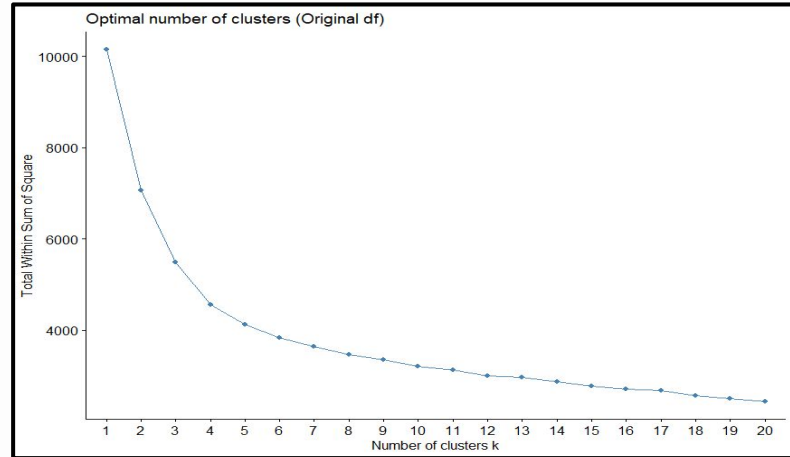


Validate PCA structure

Do the clusters align with the conclusions drawn from PCA?

Elbow and silhouette method

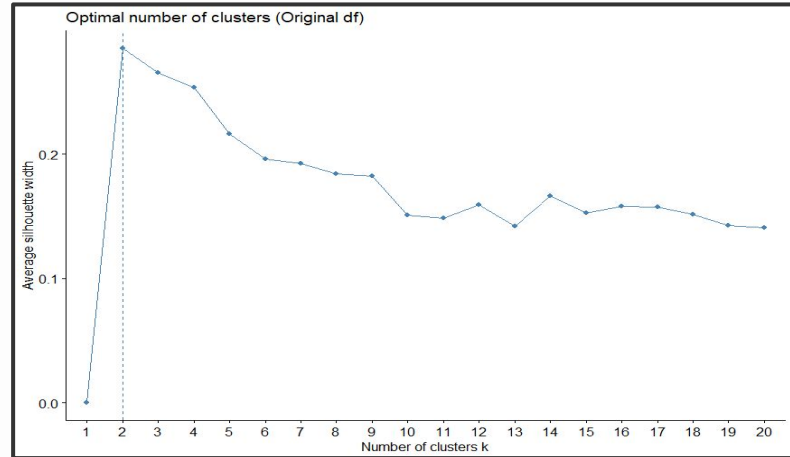
- Elbow method $\rightarrow$ $K=4$ for both datasets
- Silhouette (PCA) $\rightarrow$ confirms $K=4$
- Silhouette (original) $\rightarrow$ suggests $K=2$
- $K=2$ reflects male/female split $\rightarrow$ already known from PCA



Final decision: $K=4$

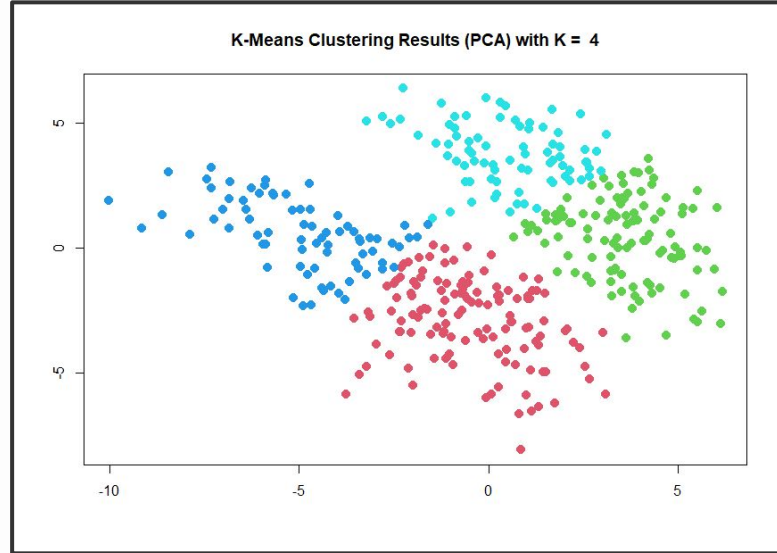
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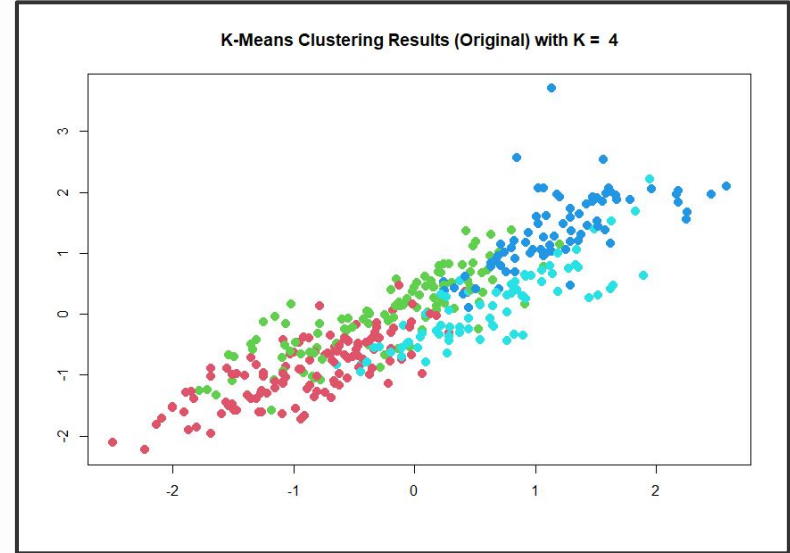


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K-means clustering

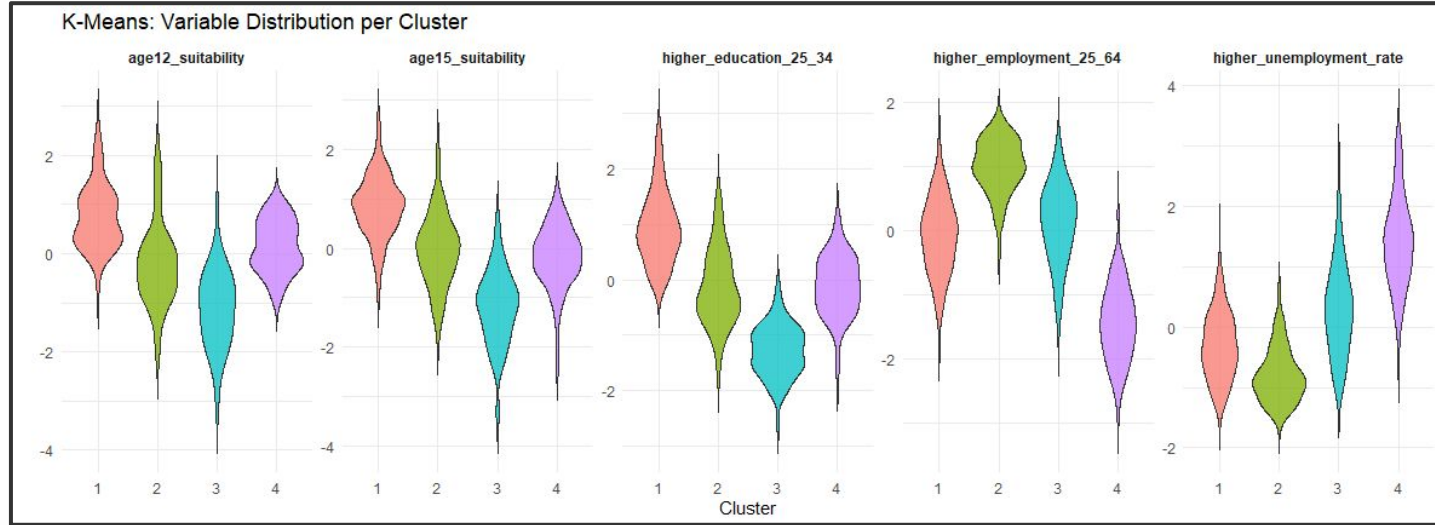


PCA clusters are well-separated, original space shows overlap along a diagonal band



With 26 correlated variables, K-Means struggles → PCA removes redundancy and exposes structure

K-means clustering



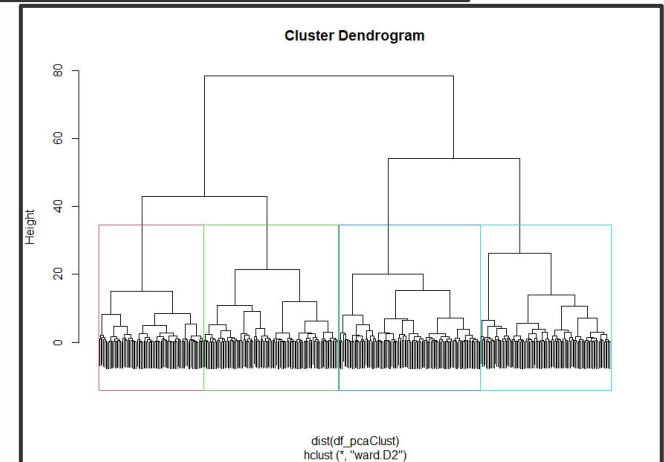
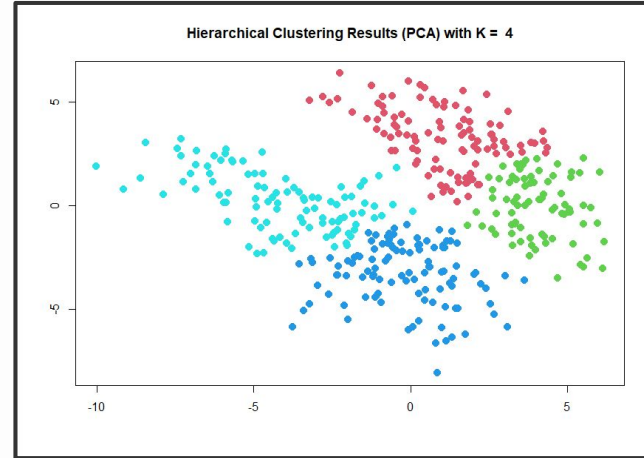
Clear socioeconomic gradient: from low education + high unemployment to high education + high employment

Age suitability variables (age12, age15) help differentiate the remaining two clusters

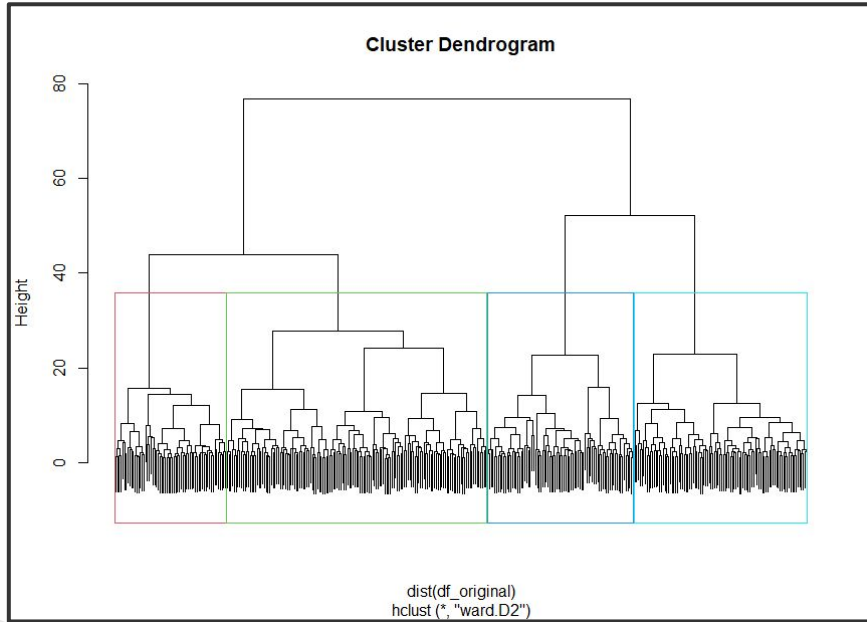
Hierarchical clustering

- Ward's D2 linkage → dendrogram confirms natural partition at K=4
- Scatter plots visually very similar to K-Means results
- Four-cluster structure appears **regardless of the algorithm chosen**

PCA dataset →

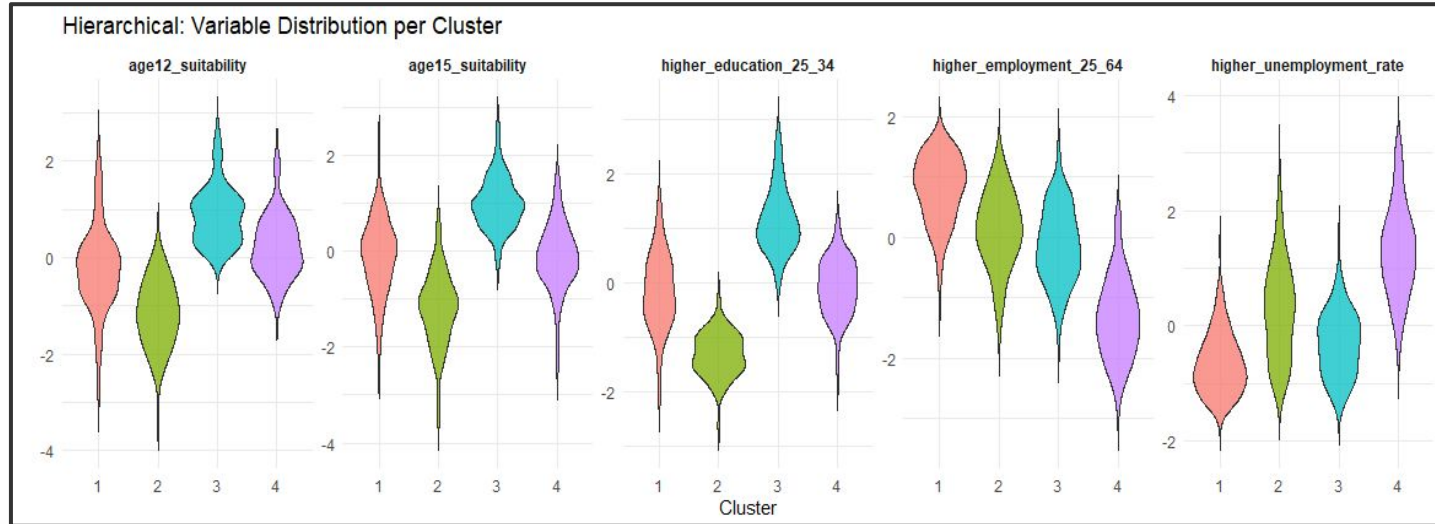


Hierarchical clustering



Original dataset

Hierarchical clustering



Same socioeconomic gradient, but tighter distributions → more internally homogeneous groups

Clusters 3 and 4 more clearly differentiated in unemployment rate and age suitability variables

Clustering conclusions

- PCA space produces more interpretable and well-separated clusters than the original space
- Four clusters reflect a spectrum from disadvantaged to high-performing Autonomous Communities
- Strong agreement between both methods → validates the four-cluster structure

06

Conclusions

Project Conclusions - Key Findings

1. Education and employment go hand in hand

-Regression quantified it, PCA structured it into two dimensions, and clustering confirmed it with a clear 4-profile gradient.

2. Strong territorial inequalities exist

-CA showed that autonomous community and unemployment are not independent; Ceuta/Melilla, Canarias, Andalucía and Extremadura consistently appear at the high-unemployment end.

3. Sex and time are structural axes

-PCA identified sex as the primary driver of variance, and the temporal trend (2013→2024) shows progressive improvement in employment and education.

Project Conclusions - Methods in agreement

1. The techniques validate each other: PCA dimensions align with the clustering gradient, and the territorial patterns from CA are consistent with the profiles found in clustering.
2. PCA was key to improving clustering — in the original space there is overlap, in PCA space the groups separate cleanly.
3. Using multiple methods gives more confidence in the results than any single one would alone.

Thanks!